An elementary polygonal proof of the outer-anticircle Bonnesen inequality

Candidate solution for fa_banach_001
agent_lane_03, model GPT5.6

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Abstract

Martini and Swanepoel asked for an elementary polygonal proof of their inequality

$$\operatorname{area}(C) + \operatorname{area}(\sigma I) \leq \sigma \operatorname{per}(C),$$

where σI is a smallest anticircle containing a planar convex body C. We give such a proof. On a common polygonal normal fan, the difference between the two sides is the quadratic area form applied to the support deficit $h_I - h_C$. Minimum circumscription supplies zero deficits whose angular gaps are at most π. An exact two-angle completion-of-squares identity makes the quadratic form nonpositive on every such gap. A common circumscribed-polygon approximation finishes the proof for arbitrary convex bodies. No mixed-area inequality is used.

1 Source request

Martini and Swanepoel, *Antinorms and Radon curves*, arXiv:math/0409200 (published in *Aequationes Mathematicae* 72 (2006), 110–138), prove in Theorem 15 that, when σI is a smallest anticircle containing C,

$$\operatorname{area}(C) + \operatorname{area}(\sigma I) \leq \sigma \operatorname{per}(C). \tag{13}$$

At the top of PDF page 19 they ask for an elementary proof involving polygons, rather than the proof through Blaschke's inequality [1].

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> Unfortunately, we do not know an elementary proof of (13) involving polygons as in the proof of (12). A proof of Blaschke's inequality (which includes (13)) may be found in [**22**], and sharpenings may be found in [**47**].

2 Idea of the Proof

Minimum circumscription forces the common supporting directions of C and I to surround the origin; two or three of them therefore cut the full normal circle into arcs of length at most π. For polygons on a common normal fan, area is an explicit quadratic form Q in the support numbers, and perimeter is its polarization against the support vector of I. Thus the desired deficit is simply $Q(h_I - h_C)$. The support deficit vanishes at the selected contacts. On each intervening arc, $-2Q$ is a discrete Dirichlet energy, and a two-angle completion of squares makes that energy nonnegative. Circumscribed polygonal approximation retains the balanced contacts and passes the result to arbitrary convex bodies.

3 Statement and normalization

Fix linear coordinates in which the given oriented area form is the standard determinant, and use the associated Euclidean scalar product only to parametrize supporting directions. Let I be the unit anticircle of a Minkowski norm. We use the elementary Steiner identity

$$\operatorname{area}(A+tI) = \operatorname{area}(A) + t\operatorname{per}(A) + t^2\operatorname{area}(I), \qquad t \geq 0. \tag{1}$$

For polygons this follows directly by adding the edge strips and corner triangles; equivalently, $\operatorname{per}(A)$ is twice the polarization of the area quadratic form evaluated at A and I. This identity is also established in the antinorm framework of the source paper [1].

Translate the smallest containing anticircle to have center 0 and dilate by $1/\sigma$. It is enough to prove the following normalized statement.

Theorem 1. *Let $C \subseteq I$, and assume that no translate of rI contains C for any $r < 1$. Then*

$$\operatorname{area}(C) + \operatorname{area}(I) \leq \operatorname{per}(C).$$

We first isolate the only consequence of minimum circumscription that will be needed. For a convex body A, write $h_A(u) = \max_{x\in A}\langle x, u\rangle$.

Lemma 1 (balanced contact normals)**.** *Under the hypotheses of Theorem 1, there are two or three unit directions u_j such that*

$$h_C(u_j) = h_I(u_j) \quad \text{and} \quad 0 \in \operatorname{conv}\left\{\frac{u_j}{h_I(u_j)}\right\}.$$

In particular, in their cyclic order, the angular gap between consecutive selected directions is at most π.

Proof. For a proposed center x, the least radius of a translate centered at x which contains C is

$$R(x) = \max_{u\in S^1} \frac{h_C(u) - \langle x, u\rangle}{h_I(u)}.$$

The hypothesis says that R has its minimum value 1 at $x = 0$. Its active directions there are exactly

$$\mathcal{A} = \{u : h_C(u) = h_I(u)\}.$$

If 0 were outside $\operatorname{conv}\{u/h_I(u) : u \in \mathcal{A}\}$, strict separation and compactness would give a vector v such that $\langle v, u\rangle/h_I(u) > 0$ uniformly on the active set. By continuity, the same strict decrease of the quotient under $x = \varepsilon v$ holds on a neighborhood U of the active set. On the compact complement $S^1 \setminus U$, the quotient at $x = 0$ is at most $1 - \eta$ for some $\eta > 0$, so it remains below 1 for sufficiently small $\varepsilon > 0$. Consequently $R(\varepsilon v) < 1$, contradicting minimality. Carathéodory's theorem in the plane now selects at most three active directions. A set of directions whose convex hull contains 0 cannot lie in an open semicircle, which is the final assertion. □

4 The discrete inequality

For $0 < \alpha < \pi$, define

$$E_\alpha(x, y) = \frac{\cos\alpha(x^2 + y^2) - 2xy}{\sin\alpha}. \tag{2}$$

Lemma 2 (discrete Dirichlet inequality)**.** *Let $\alpha_0, \ldots, \alpha_{m-1} > 0$ and $\Theta = \sum_i \alpha_i \leq \pi$. If $x_0 = x_m = 0$, then*

$$\sum_{i=0}^{m-1} E_{\alpha_i}(x_i, x_{i+1}) \geq 0.$$

Proof. For $a, b > 0$ with $a + b < \pi$, direct expansion and $\cot a + \cot b = \sin(a + b)/(\sin a \sin b)$ give the exact identity

$$\begin{aligned} &E_a(x, z) + E_b(z, y) - E_{a+b}(x, y) \\ &\quad = \frac{\sin(a+b)}{\sin a \sin b}\left(z - \frac{x \sin b + y \sin a}{\sin(a+b)}\right)^2 \geq 0. \end{aligned} \tag{3}$$

If $\Theta < \pi$, repeatedly merge adjacent intervals using (3). The sum is bounded below by $E_\Theta(x_0, x_m) = E_\Theta(0, 0) = 0$. If $\Theta = \pi$, replace every α_i by $t\alpha_i$, apply the strict case for $0 < t < 1$, and let $t \uparrow 1$. □

5 Polygonal proof

Suppose first that C and I are polygons. Take a common cyclic normal fan containing every side normal of both polygons, the contact normals from Lemma 1, and enough extra directions that adjacent angles are below π. Write

$$u_i = (\cos\theta_i, \sin\theta_i), \qquad \alpha_i = \theta_{i+1} - \theta_i \in (0, \pi),$$

with indices read cyclically, and let h_i be the support number of a polygon at u_i. Intersecting the supporting half-planes recovers the polygon; an inserted direction merely gives a possibly degenerate side.

The intersection of the two adjacent support lines gives the standard support-number area formula

$$Q(h) = \sum_i \frac{h_i h_{i+1}}{\sin\alpha_i} - \frac{1}{2}\sum_i h_i^2\big(\cot\alpha_{i-1} + \cot\alpha_i\big). \tag{4}$$

Indeed, substituting those line intersections into the shoelace formula gives (4) term by term. Thus $Q(h_A) = \mathrm{area}(A)$. Let B be the polarization of Q. Support numbers add under Minkowski addition, so (1) (or direct strip decomposition) gives

$$\mathrm{per}(C) = 2B(h_C, h_I). \tag{5}$$

Set $c_i = h_C(u_i)$, $k_i = h_I(u_i)$ and $d_i = k_i - c_i \geq 0$. Quadraticity and (5) show that the desired inequality is exactly

$$\mathrm{area}(C) + \mathrm{area}(I) - \mathrm{per}(C) = Q(c) + Q(k) - 2B(c, k) = Q(k - c) = Q(d) \leq 0. \tag{6}$$

Expanding (4) in edge form gives

$$-2Q(d) = \sum_i E_{\alpha_i}(d_i, d_{i+1}). \tag{7}$$

At each selected contact normal, $d_i = 0$. Those zero entries split the cyclic sum in (7) into paths. By Lemma 1, the total angle of each path is at most π. Lemma 2 makes every path energy nonnegative. Hence $-2Q(d) \geq 0$, proving (6) and Theorem 1 for polygons.

6 Passage to arbitrary convex bodies

Choose finite symmetric direction sets $\mathcal{U}_n \subset S^1$ whose mesh tends to zero and which all contain the selected contact directions from Lemma 1. Define simultaneous circumscribed polygons

$$C_n = \bigcap_{u \in \mathcal{U}_n} \{x : \langle x, u\rangle \leq h_C(u)\}, \qquad I_n = \bigcap_{u \in \mathcal{U}_n} \{x : \langle x, u\rangle \leq h_I(u)\}.$$

Then $C_n \subseteq I_n$, and both sequences converge in Hausdorff distance to C and I. The chosen contact constraints remain equalities for both polygons. Moreover I_n is still a minimum-radius containing homothet for C_n: if $C_n \subseteq x + rI_n$, take convex coefficients λ_j from Lemma 1 and apply the support inequalities at the selected contacts to obtain

$$1 \le r + \left\langle x, \sum_j \lambda_j \frac{u_j}{h_I(u_j)} \right\rangle = r.$$

The polygonal argument above is a purely convex statement about a pair (C, I) with I centrally symmetric: its right side is $2B(C, I)$, and the only hypotheses used are containment and minimum homothety. Since $\mathcal{U}_n$ is symmetric, I_n is centrally symmetric, so that argument applied to (C_n, I_n) gives

$$\operatorname{area}(C_n) + \operatorname{area}(I_n) \le 2B(C_n, I_n).$$

Area and its polarization are continuous under planar Hausdorff convergence. Letting $n \to \infty$ and using (1) yields

$$\operatorname{area}(C) + \operatorname{area}(I) \le 2B(C, I) = \operatorname{per}(C).$$

This proves the normalized theorem. Undoing the translation and the dilation by $1/\sigma$ gives

$$\boxed{\operatorname{area}(C) + \operatorname{area}(\sigma I) \le \sigma \operatorname{per}(C)}.$$

Novelty and scope

The theorem is classical; only the elementary proof was requested. The four run indexes, exact-title citations in the local parsed arXiv corpus, and current exact-phrase searches for the missing polygonal proof were checked on 26 August 2026. They located mixed-area proofs and papers citing arXiv:math/0409200 for other antinorm facts, but no later proof matching the argument above. This is bounded novelty evidence, not a guarantee of publication novelty.

Verification and review recommendation

The accompanying script `code/verify_discrete_energy.py` checks the merge identity, random zero-endpoint path energies, the identity $Q = -E/2$, and the support-number area formula against shoelace area. With 20,000 trials and seed 20260826 all checks pass. These computations are consistency checks, not part of the proof. Human review should concentrate on the support-number formula when redundant normals are inserted, the balanced-contact argument for translated homothets, and the simultaneous circumscribed-polygon limit.

References

[1] H. Martini and K. J. Swanepoel, *Antinorms and Radon curves*, Aequationes Math. **72** (2006), 110–138; arXiv:math/0409200, https://arxiv.org/abs/math/0409200.